CS310 Natural Language Processing
自然语言处理
Course Introduction

Instructor: Yang Xu

主讲人：徐炀

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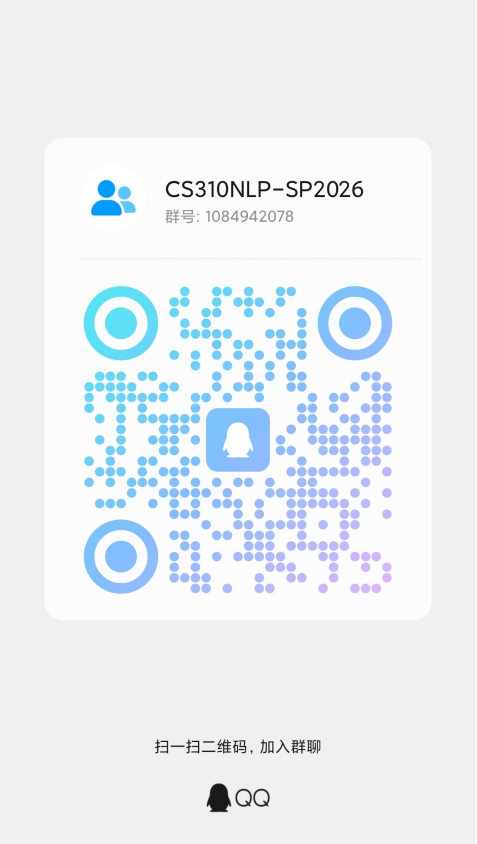
Content

- Schedule: What will be taught/learned?
- Grading Policy
- About NLP
- Programming Tools and Math

Join the QQ group:

Validation
Q: University?
A: SUSTECH



Schedule: What will be taught/learned?

Weeks	Topic	Events
Week 2	Introduction and Basic Text Processing	
Week 3	Word Vectors and Neural Networks	A1 out
Week 4	Recurrent Neural Network and Language Modeling	A2 out
Week 5	Transformer	
Week 6	Pretraining (Data, Scaling, Systems)	A3 out
Week 7,8	Post-training (RLHF, SFT, DPO)	A4 out
Week 9	Efficient Adaptation (Prompting, LoRA)	
Week 10, 11	Natural Language Generation (NLG), Benchmarking and Evaluation	Project proposal due; will have class on 5/9 (Sat.);
Week 12, 13	LLM Reasoning	A5 out
Week 14	Agent and RAG	
Week 15	Project Defense	

About Instructors

- Instructor: Yang Xu 徐炀, Associate Professor
  Dept. of Computer Science and Engineering, SUSTech
- Computational Linguistics and Consciousness Sciences Lab
  计算语言学与意识科学实验室, clcs-sustech.github.io
- Research interests:
- Computational linguistics, psycholinguistics, AI + humanity

- TA： 区煜林，陈恒威，and …

Textbooks and resources

- Book A:《大语言模型》Large Language Model，赵鑫等◎著，高等教育出版社，2024.12（2025.1重印）
- Free PDF and code: https://llmbook-zh.github.io/

- *Disclaimer: All slides and lab materials are created by instructor, with inspirations from Book A.*

Textbooks and resources

- Book B: Speech and Language Processing (3rd ed. draft), Dan Jurafsky and James H. Martin, **SLP3**
- Free PDF: https://web.stanford.edu/~jurafsky/slp3/
- *Disclaimer: All slides and lab materials are created by instructor, with inspirations from Book B.*

Speech and Language Processing (3rd ed. draft)

Dan Jurafsky and James H. Martin

Here's our Jan 6, 2026 release!

This release has is mainly a cleanup and bug-fixing release, with some updated figures for the transformer in various chapters. The August release made larger changes, including DPO in chapter 9, new ASR and TTS chapters, a restructured LLM chapter, and unicode in Chapter 2. Individual chapters and updated slides are below.

Here is a single pdf of Jan 6, 2026 book!

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Grading Policy

- Assignments: 50% (10% each)
  - Each assignment will be in mixed modality: written + coding
  - E.g., `a1_written.pdf` and `a1_coding.zip` (or .ipynb)
- Final Project: 25% (Implementation + Report + Presentation)
- Lab Practice: 20%
- In-Class Quiz: 5%

About Final Project

- 25% (Implementation + Report + Presentation)
- Group of 2 - 3 students
  - Find your partners by Week 5

- Department-wise computation support
  - Fill up the GPU application sheet:
  - For detailed instruction on the GPU-enabled cloud computing support, consult 计算机科学与工程系HPC-AI服务平台 http://172.18.34.4/

Final Project Topic Scope

- Choose from a collection of recent research papers
- Goal:
  - Replicate the original study's result.
  - Propose a direction to improve the original work (method/domain/task/…) and prove it with your own experiments.
- Grading rubrics (prone to updates)
  - Overall intellectual value and broader impact.
  - Implementation: Provide a well-documented code repo.
  - Report Writing: Clean, well-formatted, and convincing.
  - Presentation: TBD.

Labs

- 20% (Programming-based practice)
- Labs are meant to be practices (not challenging)
- Covering the lecture topic and some preparation steps for the assignments
- Bring your laptop or use the lab desktops (preinstalled with anaconda)

Learning outcomes

1. Understand the **fundamental** concepts, technological philosophy, common problems and open tasks in NLP.
2. Understand the key technology of **large language models (LLMs)**.
3. Aware of the proper technical **paths/tools** to solve common problems related to language data.
4. Can **build** pipelines to accomplish certain NLP tasks and then **evaluate** the performance.

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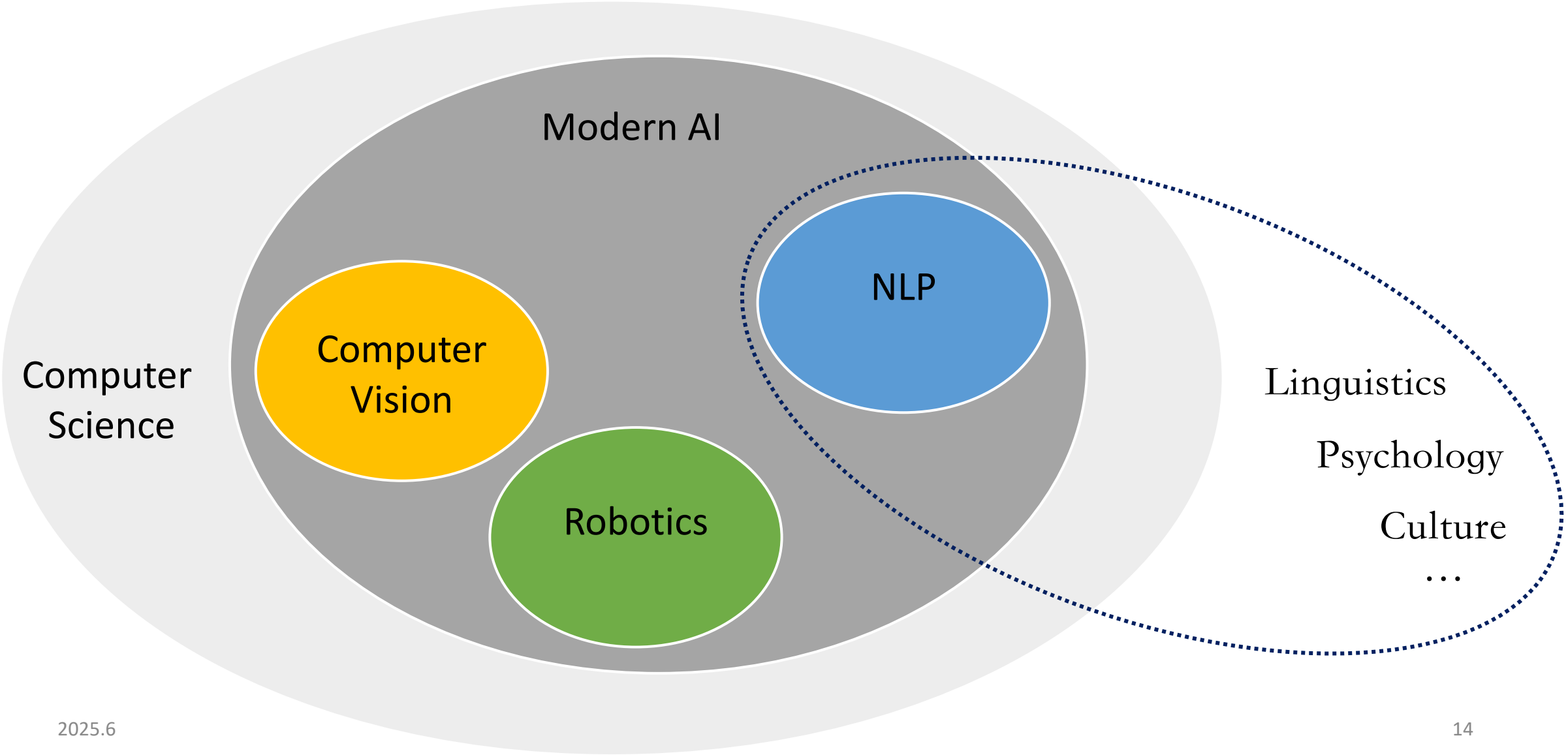
What is Natural Language Processing?



What is NLP?

- *"NLP is a branch of artificial intelligence (AI) that enables computers to **comprehend**, **generate**, and **manipulate** human language."*
  -- oracle.com

- *"NLP is an interdisciplinary subfield of computer science and linguistics … giving computers the ability to **support** and **manipulate** human language"*
  -- wikipedia.org

- *"Computational linguistics is the scientific study of language from a computational perspective. Computational linguists are interested in **providing** computational models of various kinds of linguistic **phenomena**."*
  -- aclweb.org (Association for Computational Linguistics)

Brief History of NLP

Stage 1（196X～8X）

Early NLP

Rule-based

Stage 2（199X）

Statistical NLP

- Statistics-based
- Manual feature engineering
- Labeled data
- Machine learning

Stage 3（2008～）

Neural Networks NLP

- Deep Neural-Nets
- Automatic feature extraction
- Unlabeled data

2022 onward: LLM

Brief History of NLP: Case

- A：你们家千金多大啦?
- B：千金散尽还复来
- C：千金难买少年穷

Question：Which “千金” are closer in meaning?

你们 家 千金 多大 啦?

千金 散尽 还 复 来

千金 难 买 少年 穷

Early NLP → Stat. NLP → Neur. NLP

Early NLP: Rule-based

Rule 1：

If 句子中出现和“人/human”有关的词 then “千金”代表👧

Rule 2：

if 句子中出现和“钱/money”有关的词 then “千金”代表💰

“人” 👧

A：你们 家 千金 多大 啦?

B：千金 散尽 还 复 来

C：千金 难 买 少年 穷

💰 “钱/money” “人/human”

Eearly NLP → Stat. NLP → Neur. NLP

Rule-based

💰 or 👧 ?

Early NLP: Statistics-based

Dataset： 1000 sentences containing “千金”

我们家千金大小姐

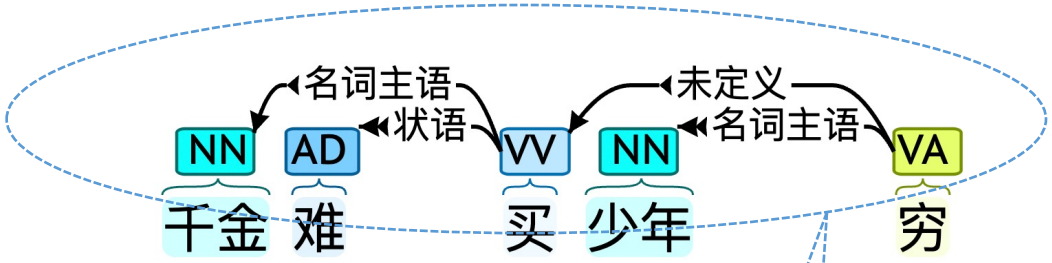
五花马，千金裘，呼儿将出换美酒

千金难买早知道

千金难买子孙贤

她可不是什么娇生惯养的千金大小姐

Learn a statistical model

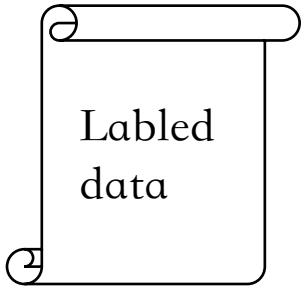


P(“千金”≈💰|feature)
= 0.9 >

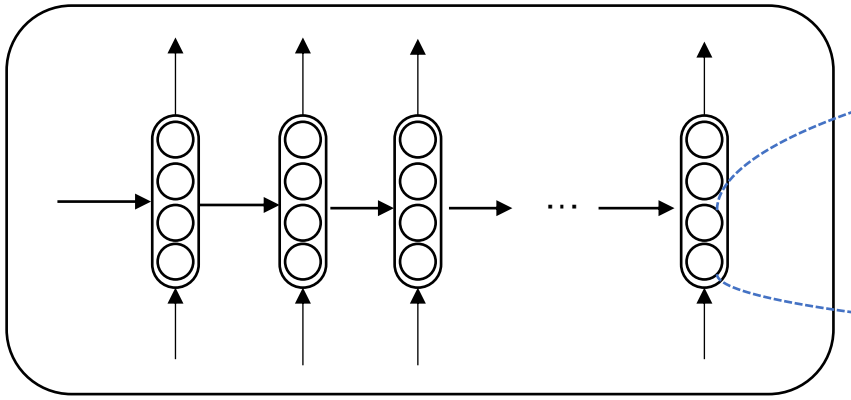
P(“千金”≈👧|feature)
= 0.1

Early NLP → **Stat. NLP** → Neur. NLP

Early NLP: Neural Network-based



Learn a neural network model



P(“千金”≈💰|auto feat.) = 0.95 >

P(“千金”≈👧|auto feat.) = 0.05

Neural-Net achieves automatic feature extraction, which is usually more effective than manual feature engineering

Early NLP → Stat. NLP → Neur. NLP

NLP in the era of LLM

- We are basically talking about Large Language Model-equipped solutions.
- LLMs that can summarize, reason, plan, execute, and evaluate.

A：你们家千金多大啦?
B：千金散尽还复来
C：千金难买少年穷

Question：Which “千金” are closer in meaning?

这是一个关于中文词语"千金"在不同语境中含义差异的趣味问题。让我分析一下：

三个"千金"的含义对比

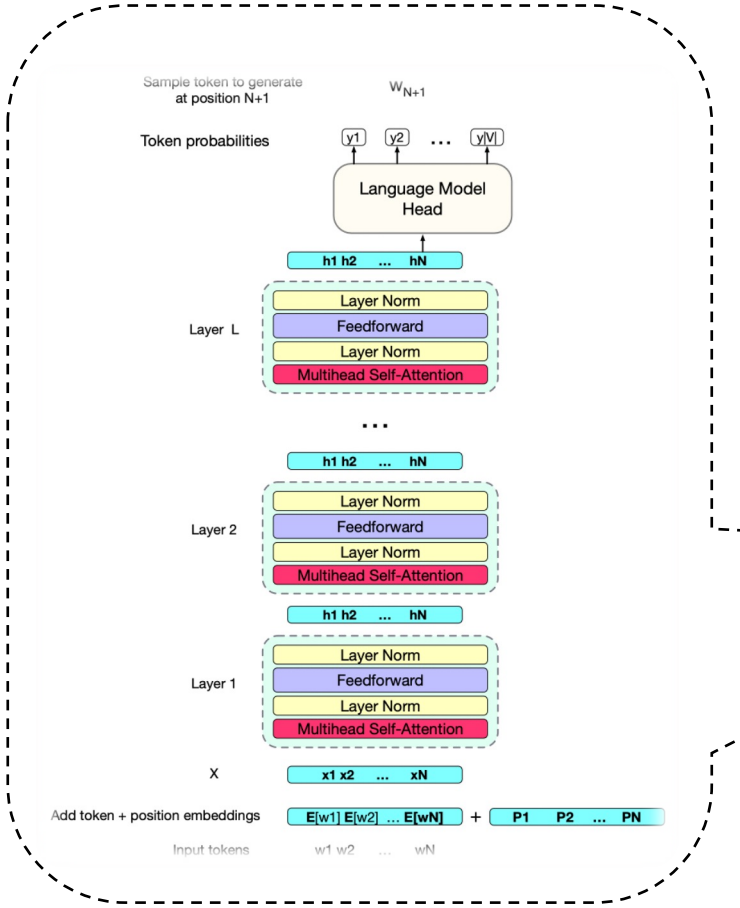
表格

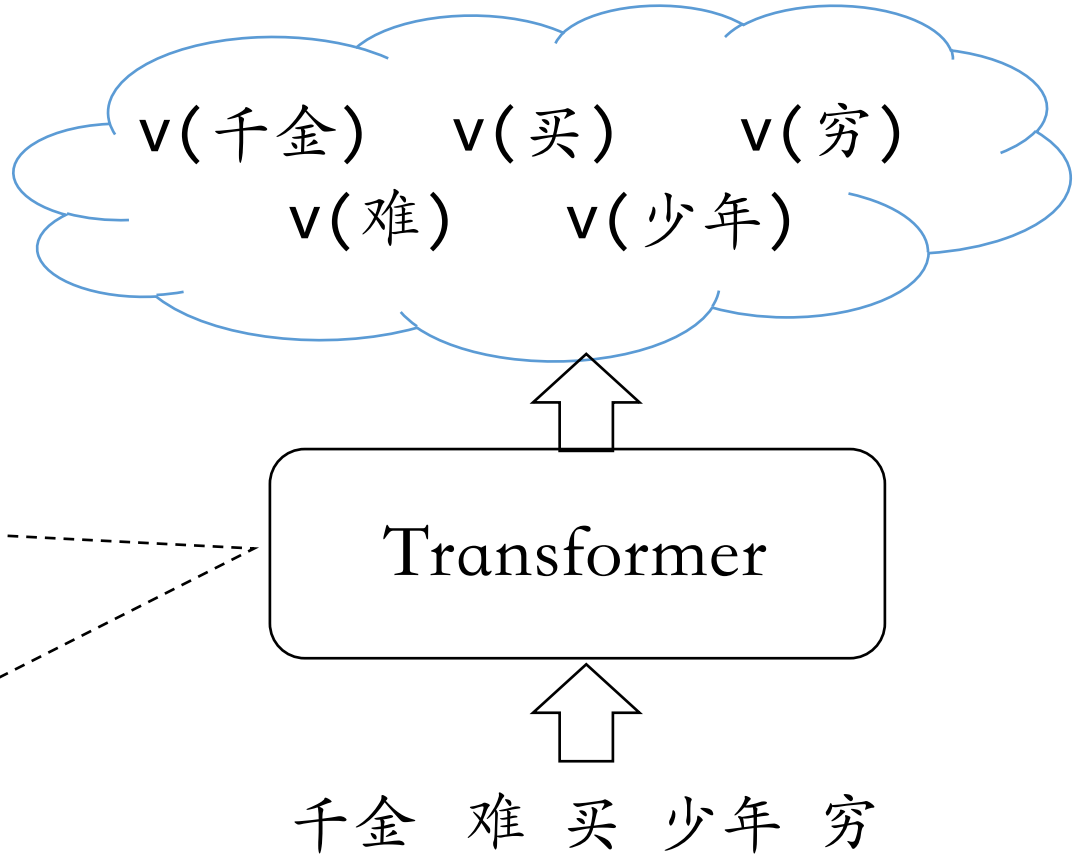
对话	"千金"的含义	语境
A	女儿（敬称）	询问对方女儿年龄，"千金"是对他人女儿的尊称
B	大量金钱/财富	引用李白诗句，意为"即使散尽钱财，也还会再回来"
C	大量金钱/财富	俗语，意为"再多的钱也买不来少年时的贫穷经历（带来的成长）"

结论

B和C的"千金"含义更接近 ✓

Break-through in Neural-Net: Transformer (~2017)

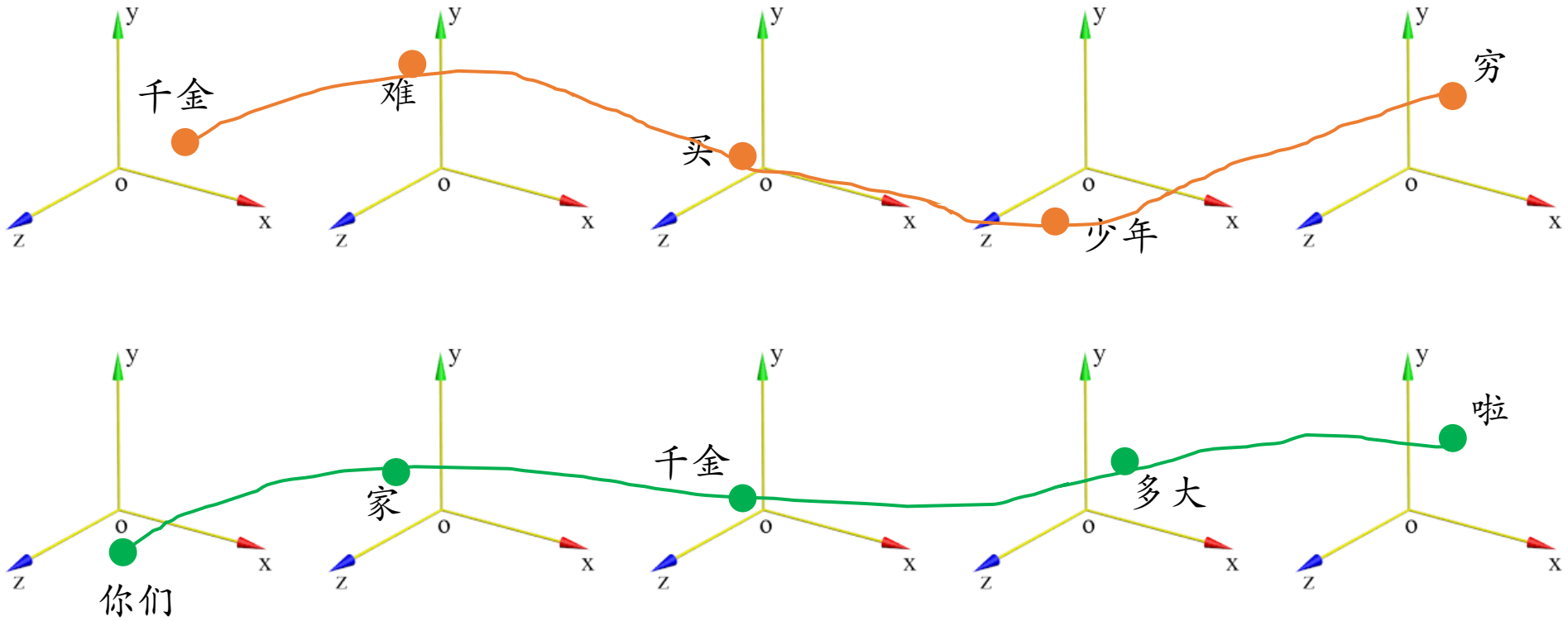




Vaswani, Ashish, et al., 2017, Attention is all you need.

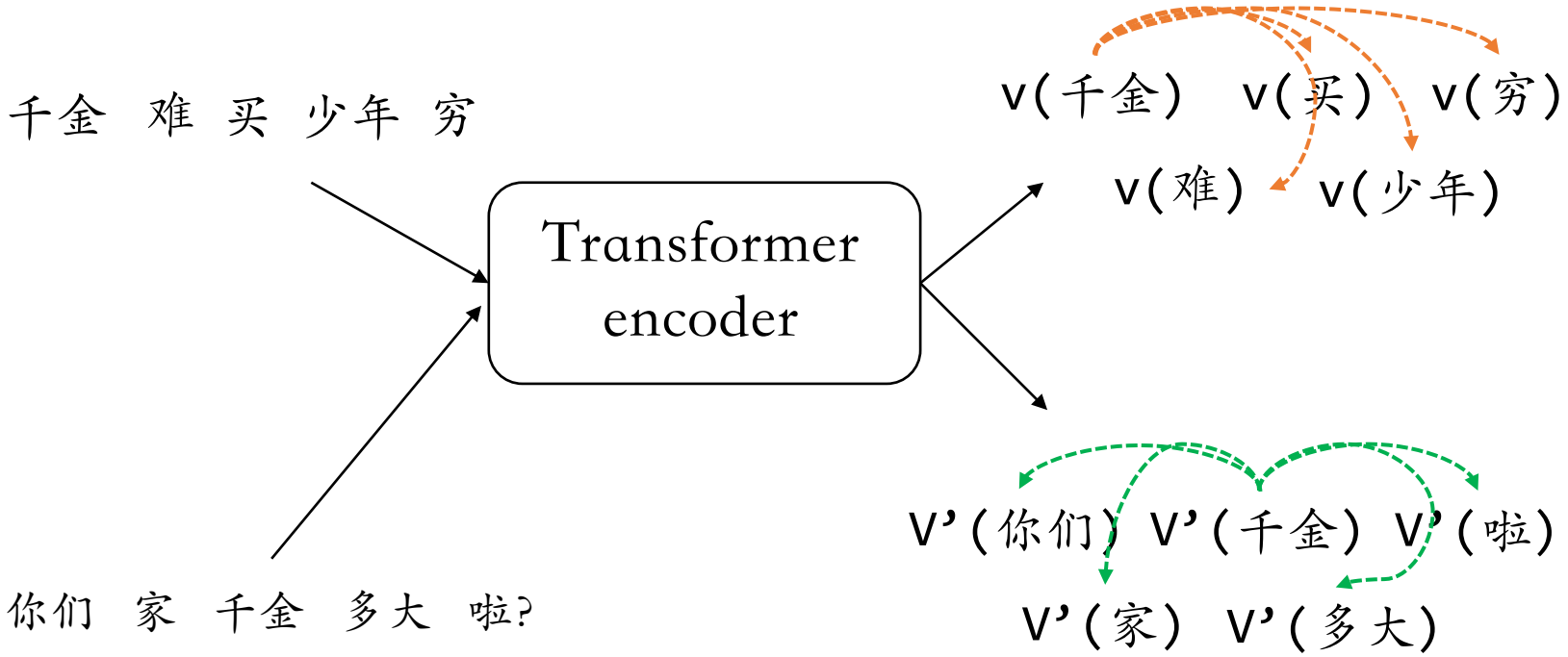
Text ⇒ Vectors in context

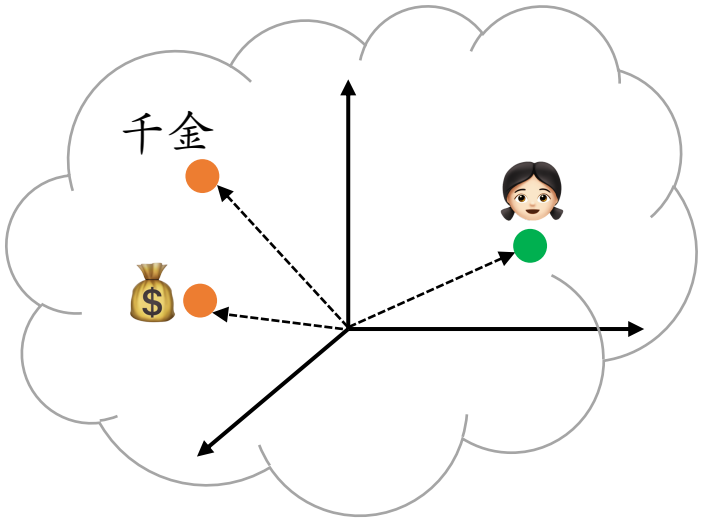
- Word/sentence vectors become trajectories in semantic space

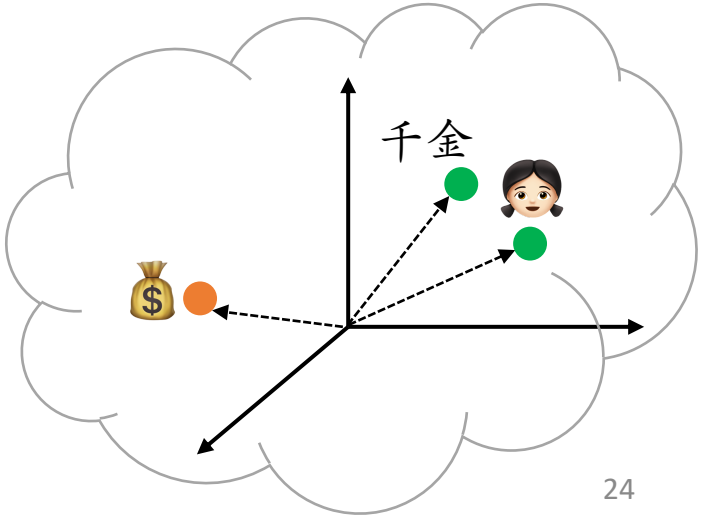


Core of transformer: Attention mechanism

- “Attention”: A smart way to vectorize words in a sentence.





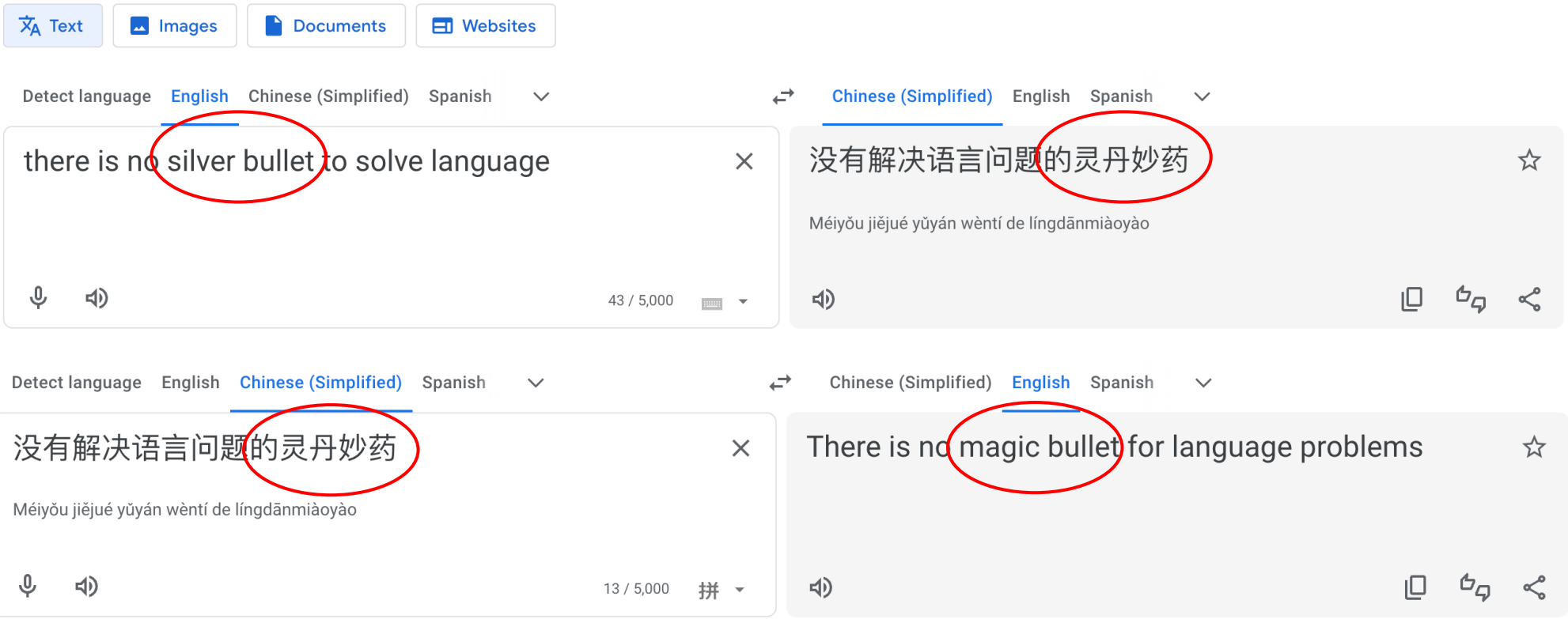


NLP in the era of LLM

- Indeed LLMs are capable of using natural languages fluently and precisely to accomplish a ton of tasks,
- But whether LLMs truly “understand” is a different question.

No silver bullet to solve language

There is always uncertainty and subjectivity in language



Some advices for learning NLP

- Be familiar with the tools (Python, neural-nets, agents, …)
- Be familiar with the fundamentals (probabilities, linear algebra, …) ; start with simple questions
- Be Willing to look into the ***raw*** data ⇒ inspiration
- Search for existing solutions before getting started
- Ask AI about any thing

- 大胆假设，小心求证
- Work on a paper when necessary

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- **Programming Tools and Math**

Python and Jupyter Notebook

- Install Python, Jupyter, PyTorch locally
- Other packages needed: NLTK (optional), numpy, PyTorch, transformers, …
- Coding agents are allowed to help you with

Math

- Calculus: derivative, chain rule, gradient descent
- Linear algebra: vector, matrix
- Probability theory
- Algorithms: dynamic programming

- Use math self-test (A0) to estimate the effort you need to put in this course.
- (A0 is not graded)

Math Self-Test

- Example

Random Diagonal

Let $\mathbf{A} = \begin{bmatrix} a_{1,1} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & a_{n,n} \end{bmatrix}$, where all entries of $\mathbf{A}$ are 0 except its diagonal entries $(a_{1,1}, \dots, a_{n,n})$.
Now suppose that each diagonal entry $a_{i,i}$ is drawn uniformly from the range $[-1, 1]$ (consider it random variable $A_{i,i}$), and assume that $n > 1$. Find:

- $p(A_{1,1} = 0)$
- $p(A_{n,n} > 0.5 \mid A_{n-1,n-1} \geq 0)$
- $p(\text{rank}(\mathbf{A}) < n)$
- $p(\mathbf{A} \succeq 0)$, i.e., the probability $\mathbf{A}$ is positive semidefinite (hint: if $n = 2$, what are the eigenvalues of $\mathbf{A}$?)

To-Do

- Read Chapter 7 of SLP3 - Neural Networks and Neural Language Models
- Complete A0 and use it to decide if you are ready to take the course